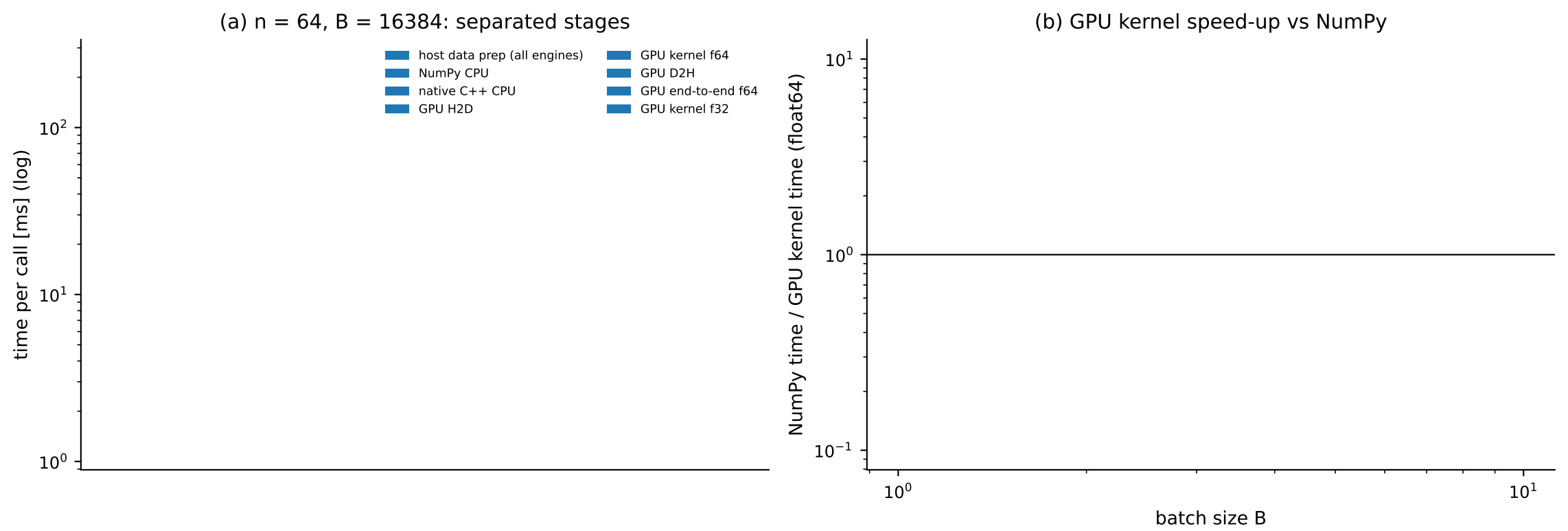
(a) n = 64, B = 16384: separated stages
host data prep (all engines)
NumPy CPU
native C++ CPU
GPU H2D
GPU kernel f64
GPU D2H
GPU end-to-end f64
GPU kernel f32
time per call [ms] (log)
10^2
10^1
10^0
(b) GPU kernel speed-up vs NumPy
NumPy time / GPU kernel time (float64)
10^1
10^0
10^{-1}
10^0
10^1
batch size B